社団法人 電子情報通信学会
THE INSTITUTE OF ELECTRONICS,
INFORMATION AND COMMUNICATION ENGINEERS

信学技報
IEICE Technical Report

ユーザ情報に基づくファッションコーディネート推薦システム

趙瑩† 荒木 健治†

†北海道大学情報科学研究科言語メディア学研究室 〒060-0804 札幌市北区北 14 条西 9 丁目
E-mail: †cecilia@media.eng.hokudai.ac.jp, †araki@media.eng.hokudai.ac.jp

あらまし 本論文はユーザの服装をコーディネートするシステムについて述べたものである．本システムはユーザの日程と使用する目的にふさわしい色やデザインを推薦することができる．日常生活の中でふさわしい服装は礼儀であり，社会的地位と個性を表現するという意味で大切なことである．我々が困るのはどの場所にどのような服装をするのがふさわしいのかということをどのようにして知るかということである．本システムはふさわしいと考えられる異なった二種類のコーディネートを選択し，帽子，上着，ズボン，靴を写真で示す．本システムはユーザのメールの内容から，季節やその日の仕事にふさわしい服装を推薦することができる．本システムで用いている情報は，季節，ユーザのスケジュール，50 種類の色を基本にして処理を行っている．実験結果より本システムがユーザがふさわしいコーディネードを得ることに有効であることが確認された．

キーワード 推薦システム、ファッションコーディネート、ユーザ情報

A Recommendation System for a Fashion Coordination based on User's Information

Ying ZHAO† Kenji ARAKI†

†Graduate School of Information Science and Technology, Hokkaido University

Kita Ku Kita 14, Nishi 9, Sapporo, 060-0804 Japan
E-mail: †cecilia@media.eng.hokudai.ac.jp, †araki@media.eng.hokudai.ac.jp

Abstract In this paper we present a recommendation system that can help users to coordinate clothes in different situations by using user's schedule and professional color matching theories. In our daily life, people need to wear properly to show their politeness, social status and personalities in various occasions, while it is not easy to decide what to wear and how to coordinate their own clothes. This system provides users two options of their fashion item coordination including hat, shirt, pants, and shoes by images. It analyzes user's e-mail information that can generate current season clothes automatically for scheduled day. The coordination results are shown as formal, casual or party clothes which were processed from over 50 color matching rules. Evaluation with several human subjects indicates that this recommendation system is helpful on choosing clothes in promising.

Keyword Recommendation system, Fashion coordination, User's information

1. INTRODUCTION

In our everyday life, we have to attend different social activities such as meeting, work, party, ceremony and so on. At the same time, a selection problem occurred very often for many people that what kind of clothes they owned and how to dress them properly for the coming events. It is not rare that these questions come to our mind sometime, for example: "What should I wear in the morning?", "Which suit should I wear for the interview?", "Does my tie match my shirt?" and "Do I look attractive in the party?" These clothes on the person showed his/her respect to the other people and his/her own taste base on the situations. If a system can store users' clothes information and select the best matching clothes for them every day, then they do not need to spend time on choosing and confuse about what to wear for the right situation. Recent years, there are more and more applications on iPhone and iPad focus on fashion coordination area which showed the numerous interests by users especially female users. On the other hand, these applications mainly have entertainment function and impressive graphics without providing users available clothes coordination for their daily use automatically. The system we proposed can automatically coordinate users' clothes and suggest two options both including hat, shirt, pants and shoes base on the situation users needed.

We used content-based recommendation method since the clothes user has means user likes most of them and do not need to use collaborative filtering [1] method that compare with other "most similar" users. Moreover, this method can avoid the "cold-start" problem by use the fashion item features to recommend users' favorite clothes.

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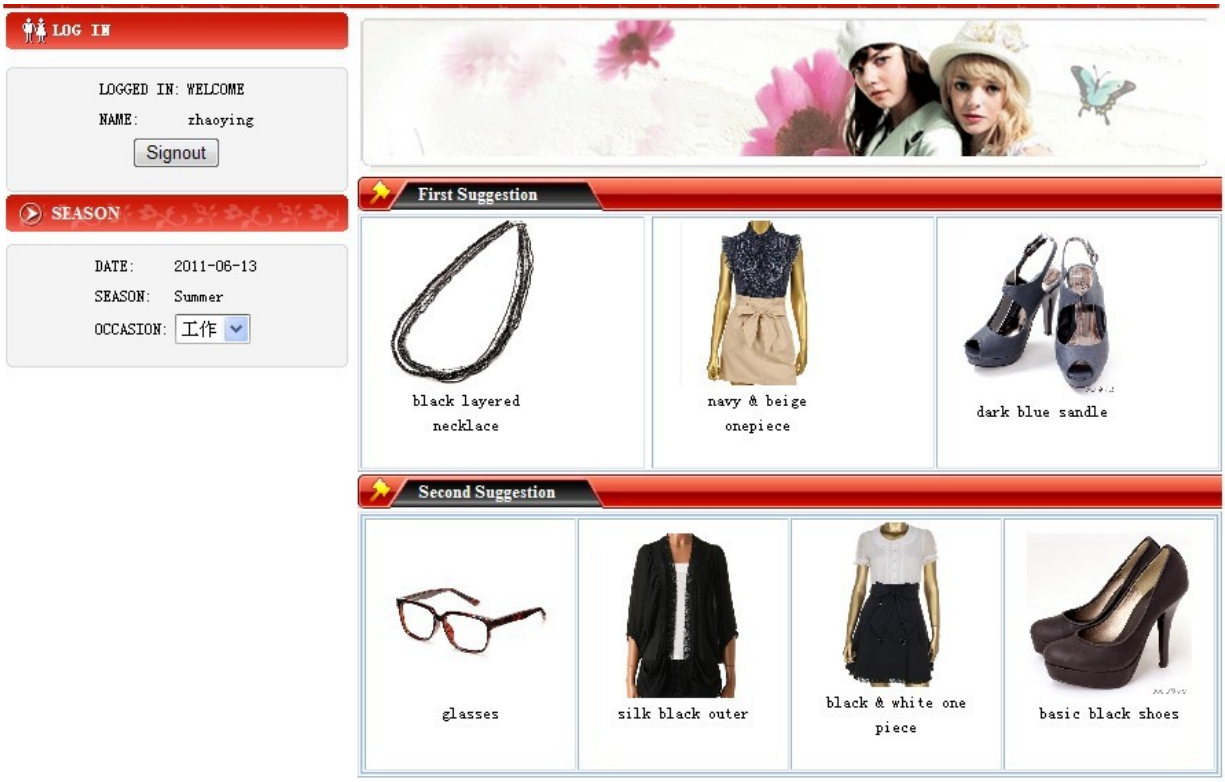


Figure 1. Example of a coordination suggestion for a female user on the situation of working in summer.

The "cold-start" problem means new users have to rate several items first, otherwise they cannot get any recommendations and new items cannot be recommended until users rate them. This system recommends fashion items base on season, situation, user's e-mail information and color matching rules. Unlike the other recommendation systems[2], one of the advantages of the system is the user's clothes data (e.g. brand, price) would not shown to other users that keeps the privacy and make people feel secure of their own taste.

2. FASHION COORDINATION SYSTEM

The goal is to build a clothes recommendation system for users to suggest what to wear automatically every day. It recommends clothes coordination base on professional color rules and by using user's personal schedule to provide suitable wear by user's information like e-mail. Besides that, the system record user's clothes wearing history that clearly for user to see and recommend other coordination next time. Content-based recommendation systems analyze item descriptions to identify items that are of particular interest to the user [3]. This system suggests two different options of clothes coordination images at the same time based on user's fashion database they uploaded. These databases divided into four seasons currently like the weather in Japan and China. When the system gives suggestions, it first find the data on season and shows them by four parts from top to bottom of the body that are hat, shirt, pants, and shoes. These parts are flexible for users to put other fashion items such as hair accessory to the hat category. Besides, female users may have a certain amount of one-piece dress, in this case the system would only show three parts of the coordination (Figure.1). The system now separate situations in three categories which are formal, casual and party (Figure.2 and 3). In order to meet users different kind of needs, a

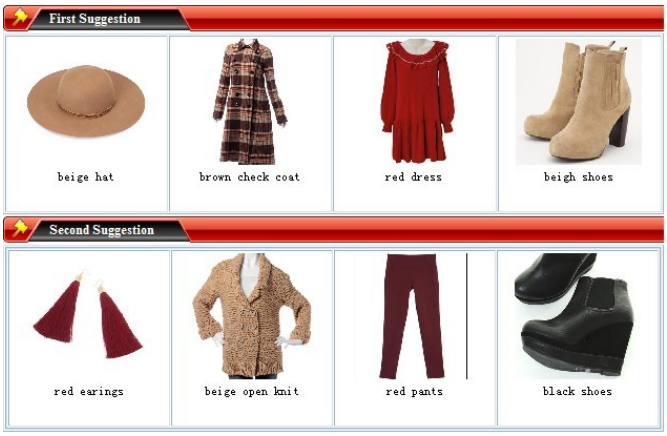


Figure 2. Example of a coordination suggestion for a female user on the situation of casual in winter.

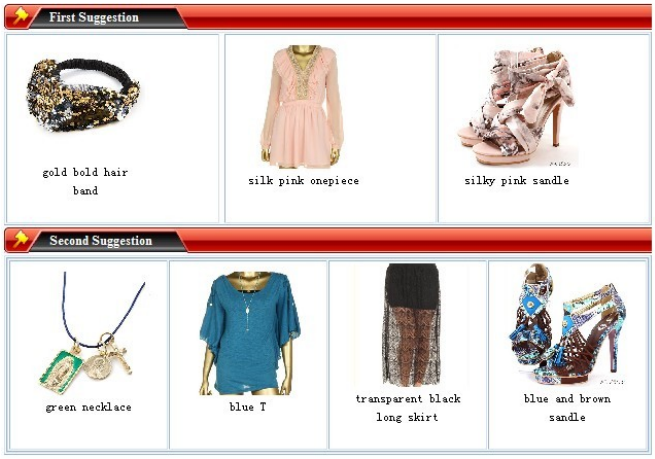


Figure 3. Example of a coordination suggestion for a female user on the situation of Party in summer.

wider range of situation categories are necessary in the future. After checking the on season clothes, the system process by checking the user's schedule. There are 2 ways to get user's schedule. The first one is user manually input the schedule from the 'Calendar' function that user can directly select a day and choose their events. The second method is the system reads user's mails and analyzes the key words in order to decide the situation automatically. At this step, the system reads all the incoming mails and separates the mails which contain key words that explain the situations like conference, meeting, party, invitation and so on. Then the chosen situation clothes would be processed by the color matching rules. There are 50 rules currently and they are from 5 basic matching types which are contrast color, similar color, black matching, brown matching, and grey matching. Since there are millions of color's name, we simply use 12 basic color type at this step. Different rules are set for different situations. In addition, these rules are gathered from professional design books for people to coordinate their clothes. After the process of reading the season, user's schedule and color matching rules, the final two results are shown on the top page. If the user does not like the coordinate, he/she can change item by click the favorite button, user can see a list of all the current season fashion items and select their favorite ones. It provides users a digital wardrobe that can manage their fashion items together and get a clear image of what they have.

3. GENERATION METHOD

This fashion coordination recommendation system makes suggestions based on three conditions which are dates and seasons, user's schedule, and color matching rules. The items are described by users own opinion that add to the comment area when user upload a new fashion item image. It contains key words such as color, material, patterns and so on. The system suggests new fashion item first since a user just bought it and it is on trend. There are several aspects we considered about building the system. They are dates and seasons, user's schedule, color matching rules, problem structure, life spam, long duration of user preference, user relationship, user input, data type and recommendation output.

3.1. DATES AND SEASONS

There are various season conditions around the world, but for this system, we set the season as the location in Japan and China. In the system, March to May is spring, June to August is summer, September to November is autumn, and December to February is winter. It is unfixable for users in other places, while it is the first version of the system. It can be manually set in the future.

3.2. USER'S SCHEDULE

In this system, user's schedule is one of the most important factors for suggestions. Since our goal is to provide suitable situation clothes with good color coordination. The user's schedule can be get from user's manually input as there is one calendar in the system for user to select date and situation. The more convenient way is the system analyzes the user's mail and record the situation on the day. It first reads the mail's title, there are key words like conference, meeting, work, business for the formal situation, party as the key word for party situation, and friend, meet, gathering for the casual situation. The mails contains schedule information would be sorted and analyze the time and date that for suggestion in the future. It is for certain that user may not have events every day, in this case, the system is giving casual wear suggestions.

3.3. COLOR MATCHING RULES

Since color matching refers to the design field, we have to summarize the various matching to practical and functional way. Therefore, we used color match books for fashion and home interiors and conclude them to 5 simple rules which contain 50 different combinations of colors. Besides, we simplified the overall colors to 12 basic colors which are red, orange, yellow, green, blue, purple, black, white, brown, grey, beige and pink. The first rule is contrast color for the situation of party. For example, the combination such as black and white, red and green,

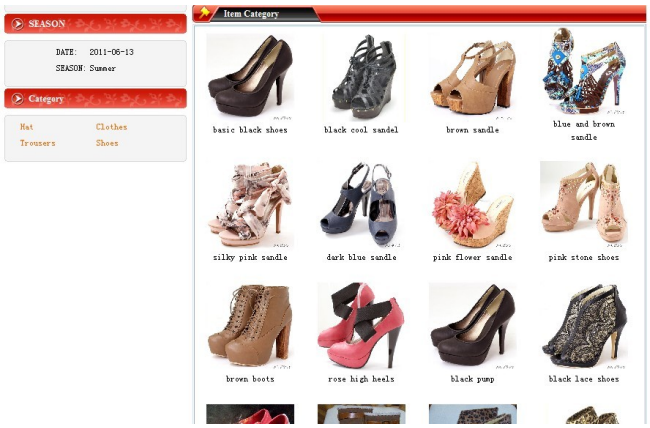


Figure 4. The category of shoes for user to check all items in four seasons

orange and blue, etc. The second rule is similar color that for the situation of casual wear, like pink and purple, beige and brown, blue and purple, beige and white, etc. In addition, there is black matching rule for the formal situation which includes coordination such as black and red, black and purple, black and brown, etc. Because in the formal situation, we have to wear black most of the time, so the system suggests black combinations for different formal days. Then, the next rule is brown matching that is for casual wear. Examples are brown and white, brown and red, brown and blue and so on. The last but not least, it is the grey matching which in the use of formal situation such as grey and purple, grey and black, grey and white, etc. We also give explanations on different coordination for user to check, for example, the explanation for the 'black and red' coordination is 'this classical combination can bring the elegant feeling'. We are going to add more color matching rules in the future for better suggestions.

3.4. PROBLEM STRUCTURE

There are four different problem structures according to [4], which are selection, configuration, planning and exploration problems. In case of choosing what clothes to wear is a selection problem. We solve this problem by providing only two suggestions to the user at one time. Unless, three or more suggestions would force the users to select again that is not helpful to avoid the problem occur. The users can check how many items they have from four fashion item categories, like shoes category see Figure.4.

3.5. LIFE SPAM

The period of time during which a recommendation is valid may be important [4]. The user habit of buying clothes, some of them do not change their fashion items

Fashion Management

ID	Fashion Name	Brand	Season	Type	Lot	Occasion	Picture	Comment
109	blue sweater	spiral girl	Spring	Clothes	First	Relaxation	image/RIMG1290.jpg	with ribbon
90	red & white skirt		Summer	Trousers	Second	Relaxation	image/RIMG1236.jpg	with heart
86	brown sweater		Autumn	Clothes	Second	Relaxation	image/RIMG0180.jpg	with gold
74	purple boots	Cecil mcbee	Spring	Shoes	First	Party	image/CEC10F0035_pz	with fur
91	grey skirt		Summer	Trousers	Second	Relaxation	image/RIMG1237.jpg	with flower
101	blue & orange top	spiral girl	Spring	Clothes	Second	Relaxation	image/RIMG1256.jpg	with flower

Figure 5. Fashion management for uploading and deleting items

for a long period while the others like young people like to buy new clothes and sell old ones frequently.

3.6. LONG DURATION OF USER PREFERENCE

Unlike the other recommendation system that has short duration of user preference. For example, a user is not interested in any suggestions after purchasing a television. As choosing clothes, people usually have consistent preference in a long duration. At the same time, what to wear is a personal style issue which means user preference is the most important factor.

3.7. USER RELATIONSHIP

In this system, it maintains long-term relationship with the user. The user has to spend more effort in order to get more accurate recommendations. Moreover, the system has a relationship with user to know their nationality, age, occupation, gender, e-mail address and so on. However, it does not have very deep relationship with the user to know user's marital status, bank account, and so forth.

3.8. USER INPUT

Every suggestion making based on the user input. In this system, the main input type is explicit type which includes the name, season, type, brand, occasion, and free comment of fashion items. Since implicit inputs are naturally noisy because they are inferred from user behavior and are thus not suitable for critical or expert domains which need explainable recommendations [2].

3.9. DATA TYPE

Users can input their fashion image data by taking photos of their clothes or save the images from web store as a lot of products we buy nowadays are easily can be found from the Internet. A various image format is

available such as JPEG, GIF, PNG and so on. Figure. 5 shows the example of fashion item database.

3.10. RECOMMENDATION OUTPUT

The system suggests two lines of fashion items. And it shows them directly and clearly. The layout designed similar to a web store because it is straightforward for user and easy to operate. An accurate output is a very important part in establishing trust between a user and system. Moreover, the suggestions output speed is within 3 seconds so that users do not need to wait for long time.

4. RECOMMEDNATION PROCEDURE

The system uses user profiles and the recommendation engine to give suggestions. A user profile describes the interests of a given user, while the recommendation engine is the computational method that computes the predictions of how much interest a given item will be to a particular user. The computation performed by the recommendation engine employs the user profiles stored in the system [3]. The user first has to register in to the system, at this step, detailed personal information needed. Next, the user can upload their fashion item images one by one and describe their color, style, occasion, function, season and so on. Thirdly, the user can input the schedule in order to generate the suitable clothes or the system reads the user's e-mail. Fourthly, the on season clothes are shown by the selection results of color matching. After that, when the user goes back to the top page, the two options of clothes coordination would be shown. Besides, the user can directly select situations from top page without input of user's schedule. If the user does not satisfied by the suggestions, he/she can directly go to the favorite list and choose their favorite item. The system records the history of suggestions, so that it would not be repeated very often.

5. EXPERIMENTS

There are two different experiment conducted to test the system. The first experiment conducted in Sapporo with 10 participants. 3 of them are males and 7 of them are females. 1 male aged 37 and 1 female aged 25 are Japanese while rests of them are Chinese. The number of their clothes database shows in Table 1. It was an offline experiment and we asked subjects to use the system in a controlled environment, they reported their experience afterwards. Users prepared their own fashion item images and start the experiment by register to the system.

In this experiment we got results from three aspects, which are system trust, system utility, and system privacy. We found out that most of users satisfied with the system and think it is functional.

TABLE I. USER AND NUMBER CLOTHES DATABASE

Users	Information		
	Nationality	***Age & Gender***	***Number of fashion items***
A	Chinese	24(Female)	138
B	Chinese	26(Female)	96
C	Japanese	37(Male)	57
D	Japanese	25(Female)	142
E	Chinese	29(Male)	52
F	Chinese	27(Female)	123
G	Chinese	27(Female)	146
H	Chinese	35(Female)	75
I	Chinese	25(Male)	70
J	Chinese	32(Female)	118

(1) SYSTEM TRUST RESULTS

A recommendation system has to build trust between users in order to give suggestions in a long period. It is a very important aspect from our point of view. The first several recommendations played an important role since it makes the first impression of how users think of the system. Therefore, we separated the subjects into two groups, group 1 (users A to E) received recommendations directly right after uploading and describing their fashion items, group 2 (users F to J) interact with the system and select their favorite items at first and then get back to receive recommendations. All of them received ten recommendations, and we found out that recommend a few items that the user already likes increase him/her trust in the system for future coordination. The results show that this system gained a certain trust well by providing users favorite items first. They rated from point 1 to point 5, and the average rates show in Table 2.

TABLE II. AVERAGE RATE BY 2 GROUPS FOR 10 RECOMMENADTIONS

Group	10 Recommendations average rate									
	1	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***
1(A-E)	2.8	3	3.2	3.6	3.6	3	3.2	3.8	3.8	3.6
2(F-J)	3.8	4	3.8	3.6	4	4	3.8	3.8	4.2	3.8

(2) SYSTEM UTILITY RESULTS

In general, we can define various types of utility functions that the recommender tries to optimize. For such recommenders, measuring the utility, or the expected utility of the recommendations may be more significant that measuring the accuracy of recommendations [6]. We measured utility in three different aspects for users. At first, how convenient the system is to the user. There are 80.0% of them think it is convenient. While 20.0% of them think the system is base on computer and it is limited. Secondly, Can the system help users to save their time on choosing clothes? There are 90.0% subjects think it is saving their time. Thirdly, the serendipity means can user find out new coordination results they have not tried before and get more interest on fashion coordination. There are 50.0% of subjects think it is interesting and can generate the coordination to change their old habits.

(3) SYSTEM PRIVACY RESULTS

Unlike the other recommendation systems that show all user's files to other users such as clothes brand, price, but we consider it is more appropriately to keep the privacy of what kind of clothes user has and choose for the day. We examined how users feel about this. 70.0% of subjects feel "very comfortable" and 30.0% of them feel "comfortable" about keeping their own clothes data and taste.

5.1. USERS' SCHEDULE EXPERIMENT

The second experiment used two people's schedule on November from 1st to 30th that testing the whole month suggestions. One participant is a Japanese female in 30s, and her occupation is office lady which means she has regular schedule working from Monday to Friday. In this case, the user set Monday to Friday as formal situation. There are 42 e-mails in total which include 9 personal e-mails for events. The system analyzed these 9 e-mails which include 6 e-mails for friends gathering, 2 e-mails for party and 1 e-mail for business trip.

By looking at the e-mails, the friends gathering are on 4th, 5th, 13th, 19th, 20th and 27th that system suggested casual wear clothes. While 2 e-mails for party invitation on 12th and 27th, system suggested party clothes successfully. There is only 1 e-mail for business trip from 21st to 25th, the system suggested formal clothes. There are 2 problems occurred on 4th and 27th that user has already set a schedule that cannot be changed.

The other participant is a Chinese male in 20s, his occupation is tour guide that has irregular schedule means he does not work from Monday to Friday. There are 23 mails in the months, and 7 e-mails contain the situations. Moreover, it can suggest accurate working situation clothes in these 7 trip e-mails. During his experiment, it was easy to give right suggestions since there are only 2 situations which are formal and the rest days are casual. We found that system can suggest proper dress base on color rules for 18 working days. Whereas, the only problem is this male user does not have many items in the winter as 17 items in total, so the same coordination suggested after several times.

They feel the system is convenient on choosing clothes and good for management of clothes. Also the system can give accurate suggestion from e-mails without user's input. It showed comfortable color matching that users satisfied.

6. CONCLUSION

In this paper, we proposed a fashion coordination system to help users save their time and get a new way to coordinate their clothes base on the schedule. The experiment also shows the system is valuable to be developed in the next step. The ideal version of this system is having color and style matching functions to give users professional recommendations.

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